



[4261] – 101

Seat No.	
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**F.E. Semester – I Examination, 2012
ENGINEERING MATHEMATICS – I
(2008 Pattern)**

Time : 3 Hours

Max. Marks : 100

- Instructions:**
- 1) Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, from Section I and Q.7 or Q.8, Q. 9 or Q.10, Q.11 or Q.12 from Section II.
 - 2) Answers to the **two** Sections should be written in **separate** books.
 - 3) **Neat** diagram must be drawn **wherever** necessary.
 - 4) Figures to the **right** indicate **full** marks.
 - 5) **Use** of logarithmic tables, slide rule, electronic pocket calculator is **allowed**.
 - 6) Assume suitable data, **if necessary**.

SECTION – I

1. A) Find the rank of matrix A by reducing it to its normal form.

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$$A = \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

- B) Examine the consistency of the system and if consistent, solve it.

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$$4x - 2y + 6z = 8$$

$$x + y - 3z = -1$$

$$15x - 3y + 9z = 21$$

- C) Verify Cayley-Hamilton theorem for the matrix A and find A^4 .

7

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

OR

P.T.O.



2. A) Find eigen values and eigen vectors for the matrix A.

7

$$A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

B) Examine whether the following vectors are linearly dependent. If so find the relation between them.

6

$$X_1 = [1 \ -1 \ 2 \ 2], X_2 = [2 \ -3 \ 4 \ -1], X_3 = [-1 \ 2 \ -2 \ 3]$$

C) Given the transformation

$$y_1 = 2x_1 + x_2 + x_3$$

$$y_2 = x_1 + x_2 + 2x_3$$

$$y_3 = x_1 - 2x_3$$

Find the co-ordinates (x_1, x_2, x_3) in X corresponding to $(1, 2, -1)$ in Y.

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3. A) Prove that :

$$\left(1 - e^{i\theta}\right)^{-1/2} + \left(1 - e^{-i\theta}\right)^{-1/2} = \left(1 + \cos \theta\right)^{1/2}.$$

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B) Find all the values of $(1+i)^{1/3}$

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C) Prove that :

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$$\text{Log tan} \left(\frac{\pi}{4} + i \frac{x}{2} \right) = i \tan^{-1}(\sinh x)$$

OR



4. A) A square lies above the real axis on an Argand's diagram and two of its adjacent vertices are the origin and the point $5 + 6i$. Find the complex numbers representing other vertices. 6

B) If $\alpha + i\beta = \tanh\left(x + i\frac{\pi}{4}\right)$, Prove that $\alpha^2 + \beta^2 = 1$. 5

C) Find the complex number Z so that, $\arg(Z + 2) = \frac{\pi}{4}$ and $\arg(Z - 2) = \frac{3\pi}{4}$. 5

5. A) Find the n^{th} order differential co-efficient for $y = \frac{x^2 + x + 1}{x^3 - 6x^2 + 11x - 6}$ 5

B) If $\sin^{-1}y = 2 \log(x + 1)$, then prove that

$$(x + 1)^2 y_{n+2} + (2n + 1)(x + 1)y_{n+1} + (n^2 + 4)y_n = 0. \quad \text{5}$$

C) Test the convergence of the series (any **one**) : 6

i) $\frac{2}{1} + \frac{2.5}{1.5} + \frac{2.5.8}{1.5.9} + \frac{2.5.8.11}{1.5.9.13} + \dots$

ii) $\frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \frac{7}{4.5.6} + \dots$

OR

6. A) Find the n^{th} order differential co-efficient for $y = \cos^4 x$. 5

B) If $y = \left(x + \sqrt{x^2 - 1}\right)^m$ show that $(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$. 5



C) Test the convergence of the following series (**any one**) :

6

i) $1 - \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} - \frac{1}{4\sqrt{4}} + \dots$

ii) $\frac{2x}{1^2} + \frac{3^2 x^2}{2^3} + \frac{4^3 x^3}{3^4} + \frac{5^4 x^4}{4^5} + \dots$

SECTION – II

7. a) Using Taylor's theorem, expand $(x-2)^4 - 3(x-2)^3 + 4(x-2) + 5$ in ascending powers of x .

5

b) Show that :

$$\log\left(\frac{1+e^{2x}}{e^x}\right) = \log^2 + \frac{x^2}{2} - \frac{x^4}{12} + \frac{x^6}{45} \dots$$

5

c) Solve (**any one**) :

6

i) Find the values of a and b such that

$$\lim_{x \rightarrow 0} \frac{x(-a \cos x + 1) + b \sin x}{x^3} = \frac{1}{3}$$

ii) Evaluate : $\lim_{x \rightarrow 0} \frac{5 \sin x - 7 \sin 2x + 3 \sin 3x}{\tan x - x}$

OR



8. a) Using Taylor's theorem, find the expansion of $\tan x$ in powers of $\left(x - \frac{\pi}{4}\right)$. 5

b) Prove that :

$$\sinh^{-1} x = x - \frac{x^3}{6} + \frac{3}{40}x^5 + \dots$$
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c) Solve (**any one**) : 6

i) Evaluate : $\lim_{x \rightarrow 0} \left(\frac{1}{x}\right)^{2 \sin x}$

ii) Evaluate : $\lim_{x \rightarrow 0} \frac{\sin^{-1} x - x}{x^3}$

9. Solve (**any two**) : 16

a) If $x = \frac{r}{2}(e^\theta + e^{-\theta})$; $y = \frac{r}{2}(e^\theta - e^{-\theta})$,

Prove that : $\left(\frac{\partial x}{\partial r}\right)_\theta = \left(\frac{\partial r}{\partial x}\right)_y$

b) If $u = \sinh^{-1} \left(\frac{x^3 + y^3}{x^2 + y^2} \right)$; Show that : $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = -\tanh^3 u$.

c) If $f(x + y + z, x^2 + y^2 + z^2) = 0$, prove that

$$(x - y) + (x - z) \frac{\partial z}{\partial y} + (z - y) \frac{\partial z}{\partial x} = 0$$

OR

10. Solve (**any two**) :

16

a) If $z = f(u, v)$ where $x = u \cos \theta - v \sin \theta$; $y = u \sin \theta + v \cos \theta$ (θ is const) then

$$\text{show that : } u \frac{\partial z}{\partial x} - v \frac{\partial z}{\partial y} = x \frac{\partial z}{\partial u} - y \frac{\partial z}{\partial v}.$$

b) If $x = e^u \tan v$; $y = e^u \sec v$ and $z = e^{-2u} f(v)$, then prove that :

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} + 2z = 0$$

c) If $2x^2 + 3y^2 + 4z^2 = 1$ and $3x - 2y + 5z = 0$, find $\frac{dy}{dx}$ and $\frac{dz}{dy}$.11. a) If $u^3 + v^3 + w^3 = x + y + z$

6

$$u^2 + v^2 + w^2 = x^3 + y^3 + z^3$$

$$u + v + w = x^2 + y^2 + z^2$$

$$\text{Show that : } \frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{(x-y)(y-z)(z-x)}{(u-v)(v-w)(w-u)}$$

b) If $u = x^2 - 2y^2$, $v = 2x^2 - y^2$

$$x = r \cos \theta, y = r \sin \theta$$

$$\text{Show that } \frac{\partial(u, v)}{\partial(r, \theta)} = 6r^3 \sin 2\theta$$

6

c) Find maximum and minimum values of $x^3 + 3xy^2 + 4 - 3x^2 - 3y^2$.

6

OR



12. a) Examine for functional dependence : $u = x + y + z$; $v = x^2 + y^2 + z^2$;
 $w = xy + yz + zx$. If dependent, find relation between them. **6**
- b) The area of triangle ABC is calculated using the formula, $\Delta = \frac{1}{2}bc \sin A$. Errors of 1%, 2%, 3% respectively are made in measuring b, c, A. If the correct value of A is 45° , find % error in the calculated value of Δ . **6**
- c) Find maximum and minimum value of $x^2 + y^2$ when $3x^2 + 4xy + 6y^2 = 140$. **6**